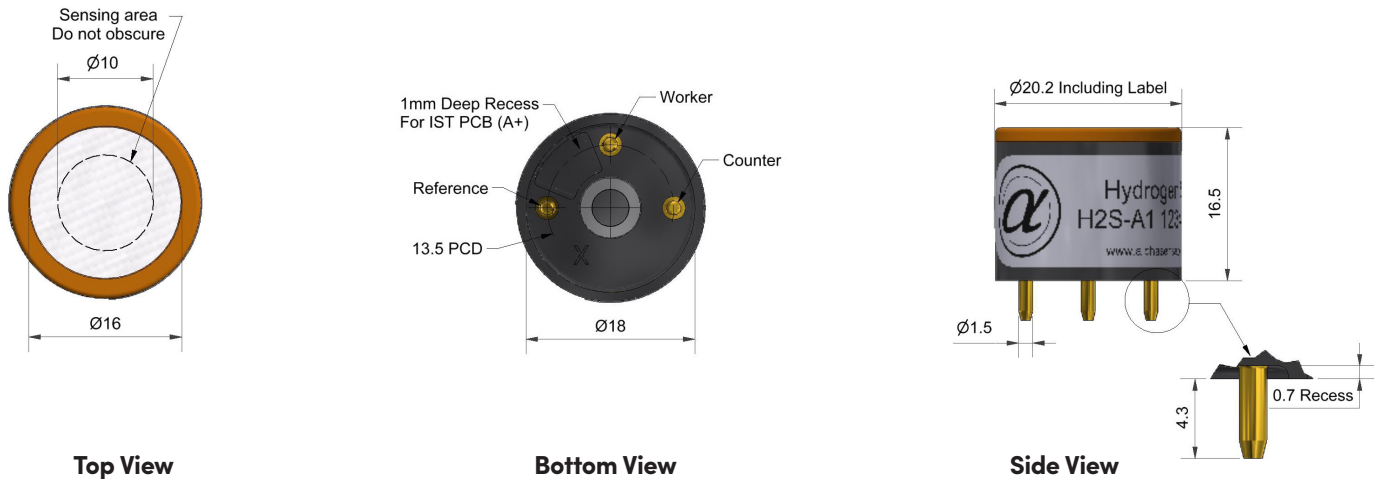


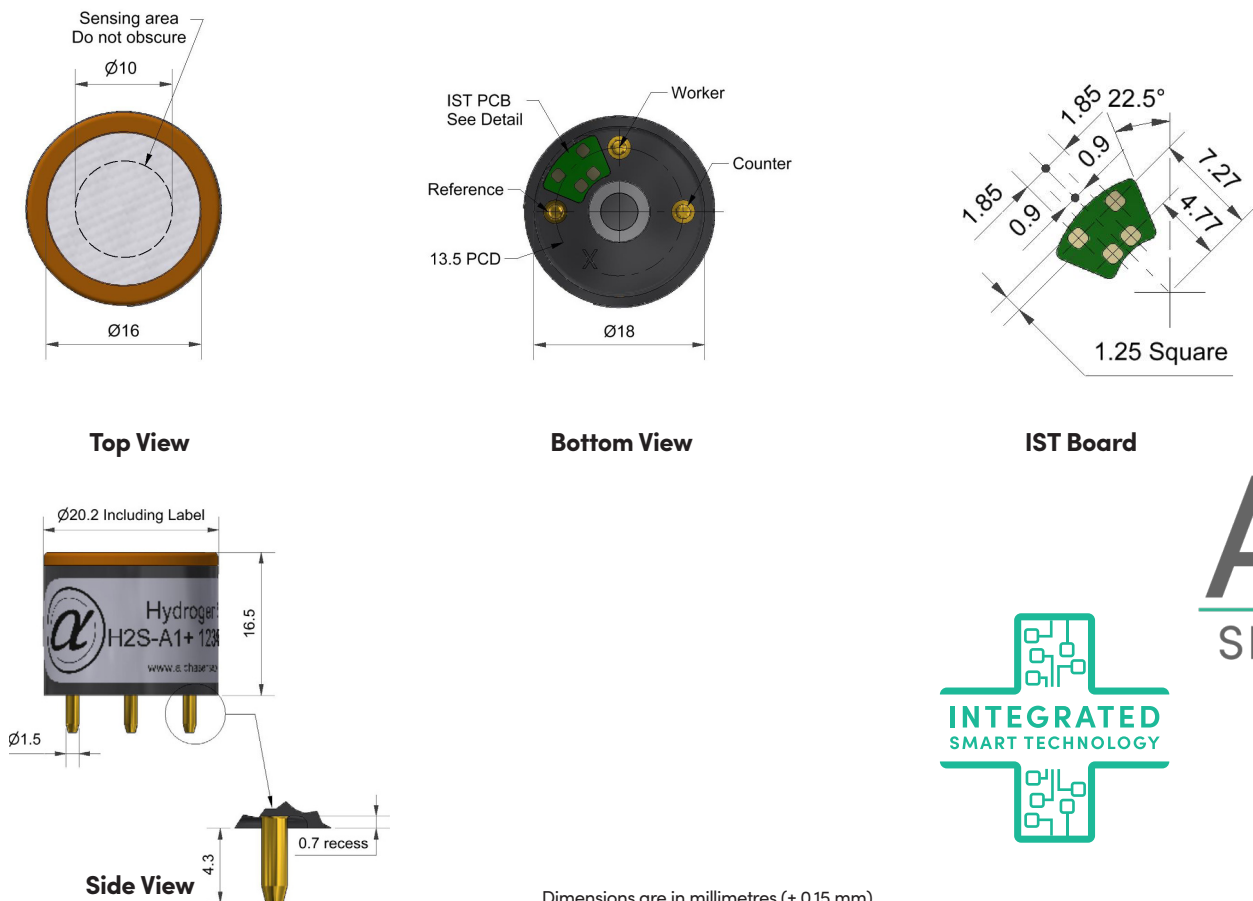
H2S-A1/H2S-A1+ A Hydrogen Sulfide Sensor

The H2S-A1 sensor is a PPM sensor that is designed for a broad variety of applications and instrumentation including portable gas detectors and fixed gas detection systems for industrial safety, environmental air quality analysis and process control. The A series is the most widely used sensor format for portable gas detection applications. It is available in our standard format (H2S-A1) and with our patented Integrated Smart Technology (H2S-A1+) that has an IST board with a memory chip and temperature sensor integrated in the sensor. The + sensors store specific calibration, specification, and identification data on every sensor allowing plug and play operation. The on-board temperature sensor improves the accuracy and simplicity of temperature compensation algorithms.

H2S-A1 Hydrogen Sulfide Sensor – 3-Electrode



H2S-A1+ Hydrogen Sulfide Sensor – 3-Electrode (with Integrated Smart Technology)



Dimensions are in millimetres (± 0.15 mm).

Sensor Data

Performance	Sensitivity	nA/ppm in 20ppm H ₂ S	550 to 900	
	Response time	t ₉₀ (s) from zero to 20ppm H ₂ S	< 35	
	Zero current (typical)	ppm equivalent in zero air	< ± 0.4	
	Resolution	RMS noise (ppm equivalent)	< 0.05	
	Range	ppm H ₂ S limit of performance warranty	100	
	Linearity	ppm error at full scale, linear at zero and 20ppm H ₂ S	0 to -4	
	Overgas limit	maximum ppm for stable response to gas pulse	500	
Lifetime	Zero drift	ppm equivalent change/year in lab air	< 0.1	
	Sensitivity drift	% change/year in lab air, monthly test	< 3	
	Operating life	months until 80% original signal (24-month warranted)	> 24	
Environmental	Sensitivity @ -20°C	% (output @ -20°C/output @ 20°C) @ 20ppm	80 to 92	
	Sensitivity @ 50°C	% (output @ 50°C/output @ 20°C) @ 20ppm	100 to 110	
	Zero @ -20°C	ppm equivalent change from 20°C	< ± 0.5	
	Zero @ 50°C	ppm equivalent change from 20°C	< ± 0.7	
Cross Sensitivity	NO ₂ sensitivity	% measured gas @ 10ppm	NO ₂	< -20
	Cl ₂ sensitivity	% measured gas @ 10ppm	Cl ₂	< -25
	NO sensitivity	% measured gas @ 50ppm	NO	< 4
	SO ₂ sensitivity	% measured gas @ 20ppm	SO ₂	< 10
	CO sensitivity	% measured gas @ 400ppm	CO	< 6
	H ₂ sensitivity	% measured gas @ 400ppm	H ₂	< 0.2
	C ₂ H ₄ sensitivity	% measured gas @ 400ppm	C ₂ H ₄	< 0.5
	NH ₃ sensitivity	% measured gas @ 20ppm	NH ₃	< 0.1
Key Specifications	Temperature range	°C	-30 to 50	
	Pressure range	kPa	80 to 120	
	Humidity range	% rh continuous	15 to 90	
	Storage period	months @ 3 to 20°C (stored in sealed pot)	6	
	Load resistor	Ω (recommended)	10 to 47	
	Weight	g	< 6	

Figure 1 Sensitivity Temperature Dependence

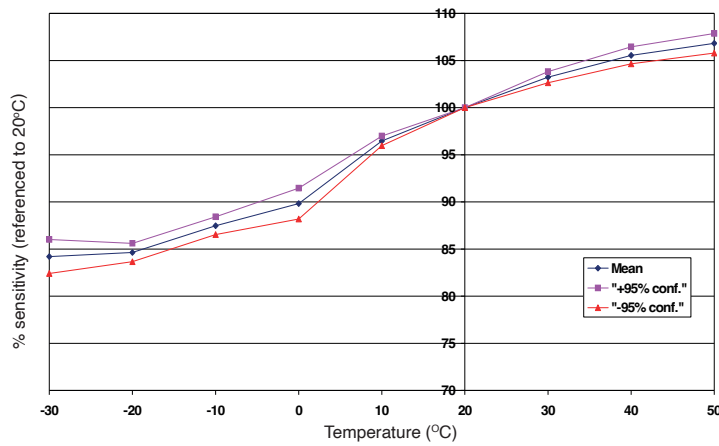


Figure 1 shows the variation in sensitivity caused by changes in temperature.

This data is taken from a typical batch of sensors.

The mean and $\pm$ 95% confidence intervals are shown.

Figure 2 Zero Temperature Dependence

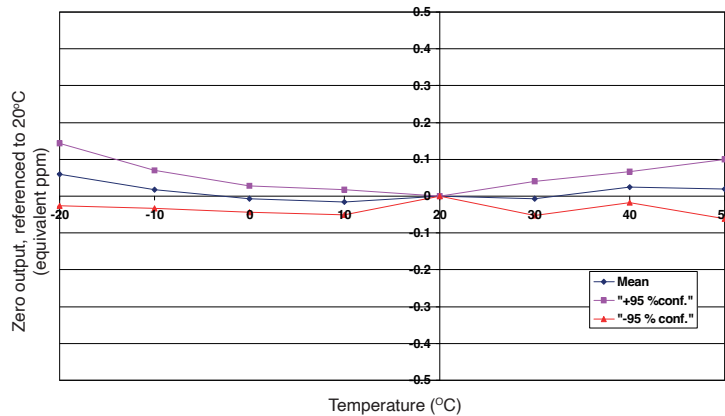


Figure 2 shows the variation in zero output caused by changes in temperature, expressed as ppm gas equivalent.

This data is taken from a typical batch of sensors.

The mean and $\pm$ 95% confidence intervals are shown.

Figure 3 Sensitivity Long-term Stability

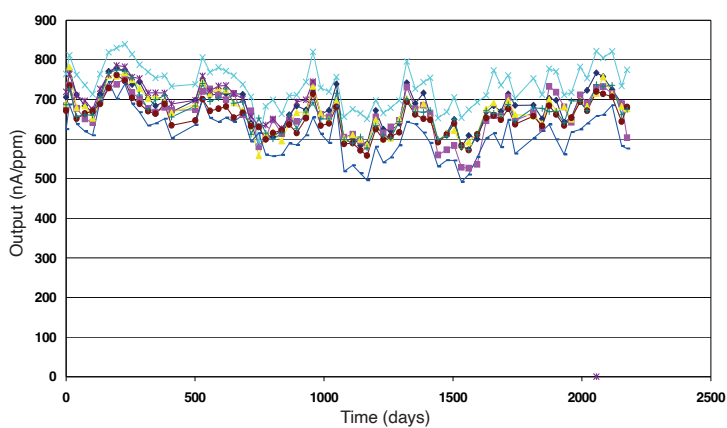


Figure 3 shows the excellent long-term stability of the H2S-A1, which results from the combination of a patented design, superior electrochemistry and good process control.

IST Board Data

Interface	Communication Bus	Compatible with the 400 kHz I ² C protocol
	Max. Bus Speed	Up to 1 MHz
	Input Logic Levels	High (Recessive) < 2.3 V Low (Dominant) < 0.2 V
	Absolute Max. Input Signal	3.6 V
Electrical	Supply Voltage Range	1.7 V to 3.6 V
	Supply current – Stand-By	< 5 µA
	Supply current – Operating	< 0.15 mA (temperature reading only) < 2.15 mA (temperature reading + memory reading/writing)
	Power Supply Conditioning	Built-In 100 nF decoupling capacitor
	ESD Protection	4 kV (human body model) – Enhanced ESD / Latch-Up protection
	Bus Pins Input Capacitance	15 pF max.
Performance	Operational Temperature	–40 °C to +85 °C
	Temperature Sensor Accuracy	±1°C (–0°C to +70°C)
	Memory Data Retention	> 200 years
	Memory Write Cycles	> 4,000,000
Data & Communication	Memory IC & I2C Address	M24128X-FCU Device Address: R – 0xA0 / W – 0xA1
	Temperature IC & I2C Address	MAX31875R0TZX+T Device Address: R – 0x90 / W – 0x91
	Product Data Start Address	0x0900
	Calibration Data Start Address	0x0B00
	User Data Area	0x0D00 – 0x18FF (3,072 Bytes)
	CRC Polynomial	0x 01 04C1 1DB7
	Digital Signature Algorithm	SHA-256

Factory-populated data

15,000+ locations

Product Data Data Format Version Customer (OEM) ID Product ID Type of Sensor / Target Gas Sensor Serial Number End of Storage Period Date Sensor Replacement Date Product Data Checksum Alphasense Digital Signature Customer Digital Signature	Calibration Calibration Data Units Zero (clean dry air) Output Calibration Span Calibration Output Sensitivity Calibration Date Calibration Data Checksum Calibration Data Signature	Sensor Specification Over-gas limit Concentration Range Temperature Range Low Temperature Range High Humidity Range Low Humidity Range High Pressure Range Low Pressure Range High Specification Checksum	Customer Specific Custom Parameters Re-Calibration Due Date Operational Limits: Low High STEL TWA Next Bump Test Due Date User Data Area
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At the end of the product's life, do not dispose of any electronic sensor, component or instrument in the domestic waste, but contact the instrument manufacturer, Alphasense or its distributor for disposal instructions. NOTE: all sensors are tested at ambient environmental conditions unless otherwise stated. As applications of use are outside our control, the information provided is given without legal responsibility. Customers should test under their own conditions, to ensure that the sensors are suitable for their own requirements.

In the interest of continued product improvement, we reserve the right to change design features and specifications without prior notification. The data contained in this document is for guidance only. Alphasense Ltd accepts no liability for any consequential losses, injury or damage resulting from the use of this document or the information contained within.(©ALPHASENSE LTD) Doc. Ref. H2S-A1/JUN26